9. LINEAR VS POLYNOMIAL

Question / Python Code

```
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import PolynomialFeatures
X = [[1],[2],[3],[4]]
y = [1,4,9,16]
l = LinearRegression()
l.fit(X,y)
p = PolynomialFeatures(2)
X_poly = p.fit_transform(X)
q = LinearRegression()
q.fit(X_poly,y)
print("Linear:", round(l.predict([[5]])[0],2))
print("Polynomial:", round(q.predict(p.transform([[5]]))[0],2))
```

Output Screenshot



Output screenshot reproduced from the supplied document.